
Network Swiss Army Knife (NSAK)

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Automating Network Security Testing



Deterministic Implementation

- › Scripted, rule-based execution
- › Reproducible results
- › Domain expertise required
- › Fast, no token cost



Agentic AI

- › Adaptive, reasoning-driven
- › Natural language interface
- › Human-in-the-loop control
- › Detailed findings reports

Evaluation: 3 LLMs × 3 Network Environments

Demo 1: Static Reconnaissance

Static Reconnaissance

Deterministic, scripted network scanning

 **Host discovery** across the network segment

 **Port scanning** and service enumeration

 **Structured findings** — reproducible, auditable output

```
nsak scenario run reconnaissance -environment home-lab
```

Demo 2: AI Reconnaissance

AI Reconnaissance

AI-driven, adaptive network analysis

- 💡 **Adaptive strategy** — AI selects the next drill at each step
- 💬 **Natural language** reasoning logged throughout
- 📊 **Coverage comparison** across Claude, GPT-4o, and Llama

```
nsak scenario run ai-reconnaissance -environment home-lab
```

Demo 3: Interactive AI Agent

Interactive AI Agent

Human-in-the-loop security automation

-  **Approval workflow** — operator confirms each drill before execution
-  **AI tool calls** — model selects, argues, and adapts its strategy
-  **Email report** — human-readable PDF findings summary

```
nsak scenario run ai-scenario -environment home-lab
```


Results & Comparison



Agentic AI

- ✓ Detailed, natural-language reports
- ✓ Low technical expertise required
- ✓ Adaptive to unknown environments
- Non-deterministic output
- Token cost, model dependency



Implementation

- ✓ Deterministic & reproducible
- ✓ Fast, no token cost
- ✓ Precise, auditable steps
- Requires domain expertise
- Findings require interpretation

*“AI excels at exploration and reporting.
Implementation excels at reliability and control.”*